Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/24/2006
PI Name:	McCready, Mark Ph.D.		
Project Title:	Determination Of Gravity Independence For Two-Phase Gas-Liquid Flows Through Conduits		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	46556- 5637	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	11/01/2004	**End Date:**	09/30/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Added to Task Book in 10/2006 per GRC		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Motil, Brian (NASA GRC) Balakotaiah, Vemuri (University of Houston)		
Grant/Contract No.:	NNC05GA23G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Work is proposed to determine quantitative criteria for gravity independence of gas-liquid conduit flows. It is recognized that for gravity independence, gravity forces must be subservient to gas and liquid inertia, surface tension and viscous forces. Relatively few studies have addressed this issue previously and no previous studies have recognized the importance of the viscous force criteria – which has already led to some discrepancies between these studies. Quantitative criteria for gravity independence would allow alternative designs for critical components that otherwise cannot be tested, full-size, in reduced and transient gravity environments. This is important because of the considerable uncertainty in quantitative and qualitative behavior of gas-liquid flows in microgravity. The work is expected to have further application for gas-liquid flows in small conduits. An example of considerable technological interest is methanol fuel cells which currently are too large to fit into cell phones and laptop computers and which further reductions in size will require overcoming problems with feeding fuel and removing waste products from small passages around the electrodes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2005; no report received during this reporting period. New to Task Book in 2006.
Bibliography Type:	Description: (Last Updated:)